TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Standardization of Potassium Permanganate with Standard Sodium Oxalate

Why permanganate can never be a primary standard, the hot oxalate titration that certifies it, the Mn^{2+} autocatalysis hiding inside the flask, and the self-indicating end point.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 7 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the exact concentration of an approximately prepared potassium permanganate solution. $KMnO_4$ is a powerful oxidising agent but a **secondary standard**: the solid always carries traces of MnO_2, and its solution is slowly reduced by dust, organic matter and light ($4\ MnO_4^- + 2\ H_2O \rightarrow 4\ MnO_2 + 3\ O_2 + 4\ OH^-$), so a weighed amount never certifies the concentration. It is made approximately (about 0.02 M — roughly 3.2 g per litre), filtered free of MnO_2, kept in a dark bottle, and standardised against **sodium oxalate**, a true primary standard: pure, stable, anhydrous, of definite molar mass ($Na_2C_2O_4$ = 134.0 g mol^{-1}). In hot dilute sulphuric acid, oxalate reduces permanganate quantitatively:

$$2\ MnO_4^- + 5\ C_2O_4^{2-} + 16\ H^+ \rightarrow 2\ Mn^{2+} + 10\ CO_2 + 8\ H_2O$$ (2 mol permanganate : 5 mol oxalate)

Three features of this reaction run the whole experiment. **Heat:** at room temperature the reaction is too slow to titrate; at 60–70 °C it is fast — but the flask must not boil, because hot oxalic acid decomposes. **Autocatalysis:** the first drops of permanganate decolourise slowly, then the reaction accelerates — the Mn^{2+} product catalyses its own formation, so the titration literally speeds itself up. **The acid must be sulphuric:** hydrochloric acid would itself be oxidised to chlorine, and nitric acid is an oxidising agent in its own right; only dilute H_2SO_4 supplies H^+ without joining the reaction.

End point: no indicator is added — permanganate is its own. The first slight excess drop colours the whole flask a **permanent pale pink**. Because the solution is so dark, the burette is read at the **upper** meniscus.

2. Apparatus and solutions

- Standard sodium oxalate solution, 0.0500 M: dissolve exactly 1.675 g of dry $Na_2C_2O_4$ (0.0500 × 0.250 × 134.0) in distilled water and make up to 250 mL in a volumetric flask.
- Approximately 0.02 M $KMnO_4$ solution in the burette (the solution under standardisation).
- Dilute sulphuric acid (about 2 M), pipette 25 mL, burette, conical flask, thermometer, water bath or hot plate, white tile.

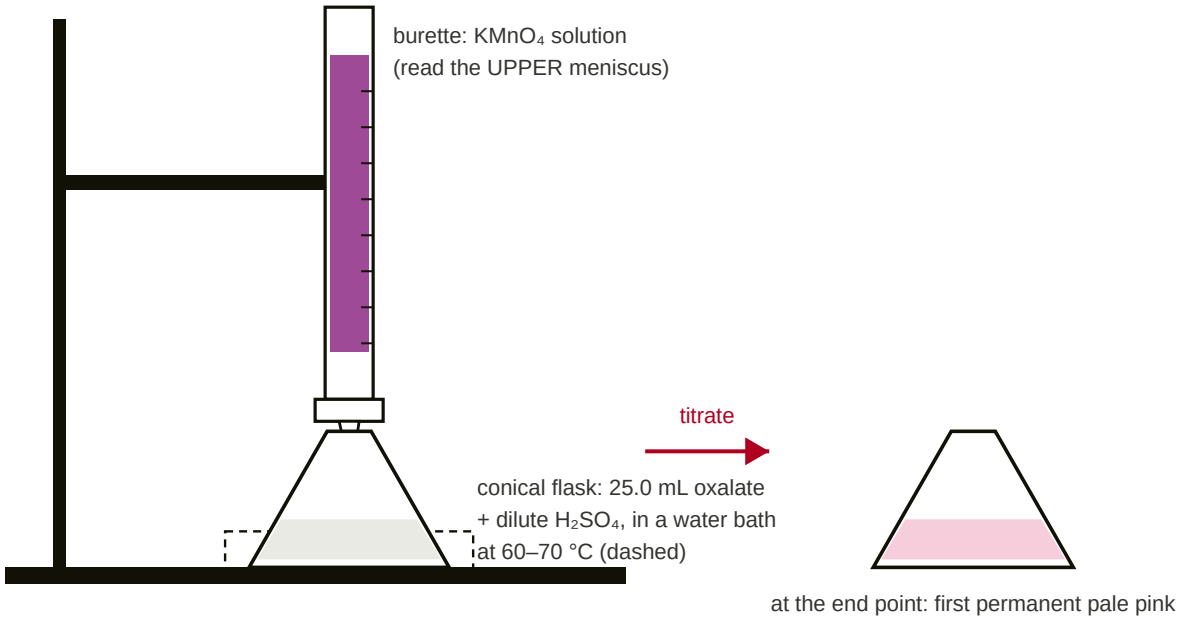


Figure 1 — Standardising permanganate against hot oxalate. The flask starts colourless; one excess drop of MnO_4^- turns the whole contents pale pink — the reagent is its own indicator.

3. Procedure

1. Rinse the burette with the $KMnO_4$ solution and fill it; remove the funnel; read the **upper** meniscus (the solution is too dark to read the lower one).
2. Pipette 25.0 mL of the standard oxalate into a conical flask; add about 20 mL of dilute sulphuric acid.
3. Heat the flask to 60–70 °C (thermometer in the liquid; do not boil).
4. Titrate with swirling. Add the first few millilitres slowly — each drop must decolourise before the next falls; once the Mn^{2+} catalyst has built up, the pink vanishes instantly and the rate can rise. Keep the liquid above about 55 °C throughout; reheat if it cools.
5. End point: the first drop that leaves a **pale pink colour persisting for 30 seconds**. Record the final reading.
6. Repeat until two titres agree within 0.1 mL; average the concordant titres.

4. Worked calculation (model data)

STANDARDISATION RESULT

Standard sodium oxalate: 0.0500 M. Volume pipetted: 25.0 mL. Concordant titre of $KMnO_4$: 25.0 mL.

Moles of oxalate = $0.0500 \times 0.0250 = 1.25 \times 10^{-3}$ mol.

Moles of $KMnO_4$ = $(2/5) \times 1.25 \times 10^{-3} = 5.00 \times 10^{-4}$ mol (the 2 : 5 ratio is the step students lose marks on).

Molarity of $KMnO_4$ = $5.00 \times 10^{-4} / 0.0250$ = **0.0200 M**.

In normality terms: oxalate's equivalent weight is 134.0/2 = 67.0, so 0.0500 M oxalate = 0.100 N; $N(KMnO_4) = 0.100 \times 25.0/25.0 = 0.100$ N, and with an n-factor of 5 in acid medium M = 0.100/5 = 0.0200 M — the same answer by the equivalents road, as it must be.

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	25.4	25.4
2	0.0	25.0	25.0
3	0.0	25.0	25.0

The rough titre locates the end point; the concordant pair does the science. Label the bottle with the standardised molarity and the date — a permanganate standardisation ages (light and dust keep reducing it) and is repeated when results matter.

6. Precautions

- Keep the titration hot (60–70 °C) but never boiling: below about 55 °C the reaction lags and the end point drifts; at boiling, oxalic acid decomposes and the titre is ruined.
- Add the first portion slowly. Drops added faster than the uncatalysed reaction can consume them pool as brown MnO_2 stains and the titration is lost.
- Never substitute HCl or HNO_3 for the sulphuric acid — one is oxidised by permanganate, the other oxidises on its own account.
- Read the upper meniscus for every reading, initial and final; mixing upper and lower readings invents a volume that was never delivered.

- Brown stains on glassware are MnO_2; they also catalyse permanganate decomposition in the burette — rinse the burette promptly after use.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is $KMnO_4$ a secondary standard, and what properties make sodium oxalate a primary one?
2. Why is the acid in this titration sulphuric acid and not hydrochloric or nitric acid?
3. The first drops of permanganate decolourise slowly, then the colour vanishes instantly. Explain — and name the phenomenon.
4. 25.0 mL of 0.0500 M sodium oxalate requires 25.6 mL of a permanganate solution. Find its molarity.
5. Why is the upper meniscus read in a permanganate burette?
6. The flask is allowed to cool to room temperature halfway through the titration. What goes wrong, and in which direction does the error push the calculated molarity?

Answer key

1. Solid $KMnO_4$ contains MnO_2 impurity and its solution is reduced by light, dust and organic matter, so neither the solid's mass nor an old solution's label states a known concentration. Sodium oxalate is obtained pure, is stable in air, anhydrous and non-hygroscopic, and has a definite molar mass — it can be weighed into an exact solution.
2. Permanganate would oxidise HCl to chlorine, consuming reagent that the calculation credits to oxalate; HNO_3 is itself an oxidising agent and would oxidise some oxalate. Dilute H_2SO_4 provides the 16 H^+ of the equation and takes no other part.
3. The reaction is catalysed by its own product, Mn^{2+} — autocatalysis. At the start no Mn^{2+} exists, so decolourisation is slow; as Mn^{2+} accumulates the reaction accelerates until it is effectively instantaneous.
4. Moles oxalate = 1.25×10^{-3}; moles $KMnO_4$ = $(2/5)(1.25 \times 10^{-3}) = 5.00 \times 10^{-4}$; $M = 5.00 \times 10^{-4}/0.0256 = 0.0195$ M.
5. The solution is too intensely coloured to see through to the bottom of the meniscus; the top edge of the liquid column is the only readable level. Consistency — upper meniscus for both readings — keeps the delivered volume correct.
6. Cold, the oxalate–permanganate reaction becomes slow: pink persists before the oxalate is truly consumed, inviting an early end point, and unreacted permanganate may fall out as MnO_2. An early (small) titre makes the calculated molarity too large (M = moles/titre).